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4WARDCRETE™

Seaford Manufacturing & Market Entry Strategy

Proprietary Products · DuPont Site Opportunity · Caribbean Export · Workforce Pipeline

SEAFORD DuPont legacy site Nanticoke River 650 acres in play	PARTY WALL AAC non-combustible 3–4 hr ASTM E119 Replaces 8" CMU	SHELTER KIT ICC 500 / FEMA P-361 Hurricane 250 mph Caribbean export	WORKFORCE L.E.E.P. Academy 400+ displaced workers Sussex County pipeline
1939 World's first nylon plant opened in Seaford	4,500 Peak DuPont / Invista employment (Seaford)	650 Acres — Invista site exploring sale (2024)	3–4 HR AAC fire rating vs. 2-hr CMU today
Merge 4 Design × Merge4Ward × M4Di Trevor Knight · Leo Lynch · Travis Davis · David Ainsworth Gwen Motley, L.E.E.P., Inc. — workforce pipeline partner		Date: June 2026 Status: Internal review — Confidential Companion: <i>ForwardGlobal_StrategyBrief_June2026.docx</i>	

1. Seaford, Delaware — The Site, the Legacy, and the Opportunity

No manufacturing location in the mid-Atlantic carries the industrial DNA that Seaford does. The DuPont Nylon Plant — now operated by INVISTA, a Koch Industries subsidiary — was built on a 650-acre site adjacent to the Nanticoke River in 1939. It was the world's first nylon manufacturing facility, designated a National Historic Chemical Landmark by the American Chemical Society in 1995. For more than five decades, it was the single most important employer in Sussex County and the economic anchor of western Delaware. And as of April 2024, INVISTA has announced it is exploring strategic alternatives — including a sale — for its nylon fibers business and the Seaford plant.

1A. The DuPont Legacy — Why This Site Matters to the Story

Ground was broken in March 1939. Nine months later, on December 12, 1939, the Seaford plant began producing nylon. In its first year of operation it ran around the clock, producing enough nylon for 64 million pairs of stockings. During World War II the plant shifted to producing military parachutes, mosquito netting, and tent material. Bulked Continuous Filament nylon — the standard material in the carpet industry — was developed at Seaford in 1958.

At its peak in the 1960s and 1970s the plant employed well over 4,500 workers — drawing labor from across the entire Delmarva Peninsula. Working at DuPont in Seaford carried a quality that is difficult to overstate for a rural county: “You could promise your son your job.” It was stable, high-paying, precision manufacturing employment in a county that had previously been almost entirely agricultural. The plant's presence fundamentally changed the economic trajectory of Sussex County.

DuPont sold the plant to Koch Industries in 2004 for \$4.4 billion as part of a larger synthetic fiber divestiture, renaming the entity INVISTA. By 2008, INVISTA restructured — laying off 400 of the plant’s 500 remaining workers — and narrowed the Seaford facility to industrial nylon staple fiber for specialty applications: military apparel, paper manufacturing press felts, and airbag fibers. By 2014 employment had dropped to approximately 400 workers. By 2021, published estimates placed the remaining workforce at closer to 100 people.

APRIL 2024 DEVELOPMENT: INVISTA publicly announced it intends to explore strategic alternatives for its nylon fibers business, including a potential sale of the Seaford facility. Barclays was engaged as exclusive financial advisor. INVISTA stated it was “too early to speculate on the final structure of a potential deal.” Mayor David Genshaw expressed optimism: “We certainly love to see that site that’s a few hundred acres being used more to employ more people. We certainly would like that for Seaford and Western Sussex.” As of June 2026, the outcome of the sale exploration is unresolved — and the site remains in play.

1B. Site Characteristics

The INVISTA Seaford facility at 25876 DuPont Road (formerly Woodland Road) covers 750 acres. The property is bordered by the Nanticoke River to the east and the Norfolk Southern Railroad to the north — with two branches of the Nanticoke along the south and southwest property lines. The EPA completed RCRA corrective action at the site; cleanup status is complete with institutional controls. An Environmental Covenant filed with Sussex County in 2009 prohibits groundwater withdrawal from designated Groundwater Management Zones and enacts land use restrictions on specific areas. The covenant is a constraint on portions of the property — but does not encumber the full 750 acres, and the industrial zoning and infrastructure of the site remain intact.

This site is not M4D’s immediate acquisition target. The INVISTA facility is a major industrial property with significant infrastructure — including a coal-fired power plant that provided on-site energy to the nylon operation — and any acquisition would involve a level of capital and due diligence that is beyond Phase 0 scope. What the INVISTA situation represents is a strategic signal: the dominant industrial employer in Sussex County is unwinding, and the community, the city government, and the state of Delaware are actively seeking a successor use that restores manufacturing employment to western Sussex. That is the opening M4D walks into.

1C. The 4WardCrete Factory — Not the Invista Site

The immediate Seaford facility target is not the 650-acre INVISTA campus — it is the available light industrial space in the city’s commercial corridor, where M4D can establish a CNC roll-forming and panel staging operation at Phase 1 capital levels.

Facility option	Size	Rate	Notes
1700 Dulaney St	40,000 SF	\$6.95/SF/yr	4 loading dock doors. Route 20 access. Active listing. City of Seaford economic incentives available. No inventory tax / no sales tax.
Western Sussex Business Campus	50+ acres (Phase 2)	TBD	Ross Station Rd & Herring Run Rd. Phase 2 design underway. Multi-tenant industrial park with infrastructure built. Suitable for build-to-suit.

Seaford Business Park	Light mfg. lots	TBD	Route 13 corridor. 75% through permit. Pad-site and warehouse options. Strong Route 13 / US 13 logistics access.
INVISTA site (partial sublease)	Negotiable	Negotiable	Long-term opportunity only. If INVISTA sale results in partial occupancy option, the existing heavy industrial infrastructure (loading docks, railroad spur access, river access) is a major asset for export operations.

Recommended Phase 1 target: 1700 Dulaney St or comparable 30,000–40,000 SF light industrial lease. This gives M4D a real operational address in Seaford, proximity to the workforce the DuPont legacy created, and the logistics access the container export model requires — without the capital burden of a large-site acquisition. The INVISTA site is the long-horizon ambition: if the sale process opens a partial or full acquisition opportunity at a price reflective of institutional controls and remediation history, Merge4Ward’s position as a ‘next-generation manufacturing’ tenant carrying the DuPont legacy forward becomes a compelling public narrative for state and county economic development support.

Port logistics: 90 miles to Port of Wilmington (primary Caribbean container service). 96 miles to Port of Baltimore (secondary). The Nanticoke River adjacent to the INVISTA site historically supported barge and light vessel traffic. Seaford’s founding connection to port access was one of the reasons DuPont chose it in 1939 — the same logic applies to container export today.

Railroad: Norfolk Southern serves the INVISTA site directly from the north property line. For bulk steel coil delivery — the primary raw material input for the 4WardCrete LGS roll-forming operation — rail delivery eliminates the cost of trucking large coil shipments from a steel mill.

Delaware advantages: No inventory tax. No sales tax. Sussex County actively recruited manufacturing through the Western Sussex Business Campus. The state’s Economic Development Office, DEDO, maintains Manufacturing Site Location incentives. The DuPont/INVISTA legacy means Seaford has a trained “light precision manufacturing” workforce base that no other Sussex County location can match.

2. Proprietary 4WardCrete™ Products — What We Make and Own

4WardCrete is not a licensed product from Framecrete or any third party. It is a Merge4Ward proprietary building system that combines standard commodity inputs — galvanized steel coil and OEM-sourced AAC panels — into a precisely engineered, tested, and documented assembly. The IP is not in the raw materials. The IP is in the engineering, the integration specification, the tested assembly documentation, and the M4Di predevelopment intelligence that prices, sequences, and delivers it.

The system produces two distinct proprietary products that address two distinct market problems:

Product 1	Product 2
4WardCrete™ Party Wall Panel A non-combustible LGS-framed AAC party wall assembly that replaces the 8" CMU party wall in US residential townhouse construction — at one-	4WardCrete™ Forward Shelter Kit A container-deployable LGS + AAC panelized building system for hurricane-resistant homes and community storm shelters. Engineered to ICC 500 / FEMA P-361

third the weight, with a higher fire rating, and without a separate masonry sub-trade. <i>US domestic market — Forward Home program upgrade</i>	standards. Made in Delaware, exported to the Caribbean. <i>Caribbean export market — Jamaica, Costa Rica, regional expansion</i>
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2A. Product 1 — 4WardCrete™ Party Wall Panel

The problem it solves: The current Forward Home system specifies an 8" CMU party wall continuous from foundation to underside of roof deck. CMU satisfies the IRC R302.2 two-hour fire separation requirement for attached townhouses. It works. But it requires a separate masonry sub-trade, a separate foundation bearing design, a separate inspection sequence, and a firestop joint detail at every CLT floor-to-CMU bearing intersection. It is the most operationally complex element of the Forward Home build sequence.

The 4WardCrete Party Wall Panel: A 100mm to 125mm (4"–5") AAC panel set within the LGS framing system. Non-combustible per ASTM E136. Fire-rated per ASTM E119. Documented under AERCON Florida ESR-1652.

Attribute	8" CMU (current)	4WardCrete Party Wall Panel
Fire rating	2-hr ASTM E119 (inherent)	3-hr @ 100mm 4-hr @ 125mm ASTM E119 (inherent)
Non-combustibility	✓ ASTM E136	✓ ASTM E136
Weight per SF	~80 lbs/ft² (8" CMU)	~27–32 lbs/ft² (~35% of CMU)
STC acoustic rating	≥50 dB inherent	40–45 dB inherent 50+ dB with resilient channel on one face
Installation trade	Separate masonry sub-contractor Separate inspection sequence	Same LGS framing crew No separate trade required
Bearing element	CMU bears CLT floor loads at every level — firestop joint detail at each intersection	LGS frame carries all loads AAC is non-structural enclosure No bearing intersection detail
ESR / documentation	Inherent code compliance — no ESR required	AERCON Florida ESR-1652 (ICC-ES — all I-Codes)
Code path	IBC Table 722.5.2 / IRC R302.2	IBC Table 722.5.2 / IRC R302.2 via ESR-1652 tested assembly
Impact on AE24-002	N/A — CMU is current spec, AE24-002 cannot be changed	Forward specification for Type A/B projects post-Claymont G4

CRITICAL SPEC NOTES — PARTY WALL: 1. AE24-002 Claymont is already engineered with CMU and cannot be changed. 4WardCrete Party Wall is a forward specification only. 2. Reinforcing steel inside structural AAC panels must be epoxy-coated in all coastal and tropical deployments. 3. STC ≥50 requires resilient channel on at least one face of the AAC panel assembly. 4. The AAC party wall panel requires exterior cladding or render in any application with direct weather exposure.

2B. Product 2 — 4WardCrete™ Forward Shelter Kit

The problem it solves: Jamaica’s post-Hurricane Melissa reconstruction baseline is now Category 5 specification. More than 90,000 homes were destroyed or severely damaged by a storm that made landfall at 185 mph sustained winds. NaRRA’s explicit mandate is ‘build back better’ — meaning every replacement home must exceed the standard of what was destroyed. The existing Jamaican housing stock — largely unreinforced concrete block construction — failed catastrophically. The next solution must be engineered, documented, rapid-to-deploy, and culturally legible to Jamaican homeowners and code officials who know concrete construction.

The 4WardCrete Forward Shelter Kit: A container-deployable assembly of pre-cut LGS framing (CNC fabricated at Seaford), OEM-sourced AAC enclosure panels, and Allura fiber cement cladding — complete with installation drawings, hold-down schedule, and M4Di ALDi Tracker QR code. One 40-ft container carries the complete framing and enclosure package for one standard 2-bedroom unit. Kits arrive at the site numbered, labeled, and sequenced. Installation requires basic LGS framing skills — trainable in days, not years.

2B-i. ICC 500 / FEMA P-361 Hurricane Shelter Pathway

This is the product’s most strategic differentiation — and it opens a US domestic government procurement channel that no other element of the Forward System touches.

ICC 500-2023 is the American National Standard (ANSI-approved) for the design and construction of storm shelters. It is incorporated by reference in the 2009 through 2024 editions of the International Building Code (IBC) and the International Residential Code (IRC). FEMA P-361 adds the funding criteria that make a storm shelter eligible for FEMA Hazard Mitigation Grant Program (HMGP) funding. FEMA P-320 provides prescriptive plans for residential safe rooms that meet both standards.

Standard	What it requires — and how 4WardCrete meets it
ICC 500-2023 (minimum design standard)	Resistance to extreme wind and wind-borne debris. Design wind speeds of 250 mph for residential tornado shelters; 130–175 mph for hurricane shelters depending on location. LGS framing with engineered hold-downs, CFS shear walls, and AAC enclosure can be designed to meet 250 mph wind loads per ASCE 7. The combination of inorganic AAC (no combustibility) and LGS (no decay) satisfies the durability requirements. Impact-rated glazing or shutters at all openings is a separate line item in the shelter kit.
FEMA P-361 (safe room funding criteria)	All ICC 500 requirements plus: fire-resistant rated construction, minimum 8" of concrete or equivalent for exterior walls, independent peer review required, and specific siting requirements. The ‘equivalent to 8" concrete’ standard is met by reinforced AAC at 200mm (8") with documented test data. 4WardCrete community shelter modules specify 200mm reinforced AAC with stainless rebar + LGS frame. Independent structural peer review is a standard M4Di deliverable.
FEMA HMGP (Hazard Mitigation Grant Program)	Structures meeting FEMA P-361 criteria are eligible for HMGP grant funding. This means a 4WardCrete community shelter built in a US coastal community can be grant-funded by FEMA — a government procurement channel completely separate from the Caribbean export market. Every hurricane season opens new HMGP funding cycles. Delaware, Virginia, Maryland, and North Carolina coastal communities are active HMGP applicants.
CUBiC (Caribbean Uniform Building Code)	CUBiC explicitly recognizes LGS framing for hurricane-force wind resistance and is based on ASTM standards — the same standards AERCON ESR-1652 and AISI S240 are written to. The Forward Shelter Kit specification document cites both CUBiC and ICC 500 compliance, making it immediately legible to Jamaican code officials and to NaRRA’s ‘build back better’ procurement requirements.

ICC 500 + FEMA P-361 = TWO MARKETS IN ONE PRODUCT: The 4WardCrete Forward Shelter Kit is simultaneously a Caribbean export product for Jamaica reconstruction and a US domestic government procurement product for coastal community storm shelters eligible for FEMA Hazard Mitigation Grant funding. The same container kit — LGS + 200mm reinforced AAC + impact-rated openings — satisfies both. Seaford manufactures one product that competes in two markets.

2B-ii. Forward Shelter Kit — Container Specification

Each 40-ft container export kit contains the complete assembly package for one standard 2-bedroom Forward Shelter Kit unit. The container is the delivery vehicle and the inventory management system simultaneously:

Kit component	Specification
LGS framing package	Pre-cut, pre-punched CNC cold-formed steel framing for one unit. G235 galvanizing (ASTM A653) for HVHZ and marine coastal environments. Bundled by assembly sequence: foundation tracks → wall studs → headers → trusses → hold-down connectors. Labeled numerically with matching installation drawing reference.
AAC enclosure panels	AERCON or XELLA/Ytong OEM panels. Standard units: 100mm for non-structural exterior walls. 200mm reinforced AAC for ICC 500 shelter specification. Pre-cut to unit dimensions at Seaford staging area. Pre-labeled with orientation and assembly sequence.
Impact-rated openings	Impact-rated window units (DP50 minimum per FBC HVHZ) and door systems meeting ICC 500 debris impact requirements. Pre-glazed where possible. Included in kit for ICC 500 compliance.
Allura cladding package	Allura Panel fiber cement cladding, pre-cut to unit elevation dimensions. Non-combustible. Moisture and salt resistant. 50-year warranty. Mandatory for all tropical deployments — prevents AAC moisture absorption in high-humidity environments.
Mechanical rough-in package	PEX plumbing, wiring harness, and HVAC stub-out kit matched to destination climate (mini-split spec for Caribbean). Optional upgrade to full T2–T4 Forward Home MEP spec.
M4Di documentation	Installation sequence drawings. Hold-down schedule with torque specs. ALDi Tracker QR code linking to project-specific G0 ROM. JNBC / CUBiC compliance statement. AERCON ESR-1652 and AISI S240 reference documents.
Container weight	One 40-ft standard container carries framing + panels + cladding for one 2BR unit. AAC is ~35% the weight of equivalent CMU. LGS framing is ~10–15% the weight of equivalent structural steel. Total kit weight well within standard container payload limits.

2C. What the Seaford Factory Actually Does

The factory is not an AAC manufacturer. Autoclaving AAC panels requires a pressure vessel approximately 3 meters in diameter and 45 meters long — \$2M+ capital investment. The Seaford facility is an LGS fabrication and kit integration operation:

Operation	What happens
Steel coil receiving	Hot-dipped galvanized steel coil arrives from US steel mill by truck or rail (Norfolk Southern spur at INVISTA site / truck to Dulaney St.). G90 for US domestic kits. G235 for Caribbean export. Coils stored on-site.
CNC roll-forming	Steel coil feeds into CNC roll-forming machine. Software loads the digital model file for the specific unit type. Machine cuts, punches, and forms studs, tracks, headers, and trusses to ±1mm precision from 3D model. No field cutting required at destination. One CNC line produces framing for 3–5 housing units per shift.
AAC panel receiving & staging	Pre-manufactured AAC panels arrive from AERCON Florida by truck or from Caribbean regional supplier by container. Panels staged at Seaford by lot, trimmed to unit dimensions as needed, labeled with assembly sequence codes.
Kit assembly & containerization	One team assembles the complete container kit: LGS package + AAC panels + Allura cladding + mechanical rough-in + documentation package. Container is sealed, weighed, and manifest prepared for export via Port of Wilmington.
Forward Home SIP staging (dual-use)	The same facility receives and stages 4Ward-SIB SIP panels from Barricade/INDEVCO for the US domestic Forward Home program. Panels sequenced by lot address and dispatched by truck to Wilmington job sites. Shared overhead between two product lines improves factory economics from Day 1.
M4Di ALDi QR labeling	Every kit component is labeled with a QR code that links to the project-specific M4Di ALDi Tracker entry. Installation crews in Jamaica or Wilmington scan the label to access assembly drawings, hold-down schedules, and inspection checkpoints. This is the digital intelligence layer that differentiates 4WardCrete from Chinese prefab and conventional construction.

CAPITAL REQUIREMENT — PHASE 1 SEAFORD FACTORY: Facility lease (30,000–40,000 SF at \$6.95/SF/yr): \$140K–\$210K/yr. CNC roll-forming line: \$150K–\$250K (production-grade machine, new or refurbished). AAC panel staging equipment (forklifts, cutting station): \$40K–\$80K. Tenant improvements, 3-phase power upgrade, loading dock: \$50K–\$100K. TOTAL Phase 1 capital: \$200K–\$400K. No autoclave. No AAC manufacturing. This is the entry point — not the full build-out.

3. Workforce Training & Employment — The Sussex County Opportunity

The workforce narrative is not a corporate social responsibility add-on. It is the foundational reason the 4WardCrete manufacturing strategy is positioned in Seaford rather than anywhere else in Delaware. The DuPont/INVISTA legacy created a community in which four generations of families built their economic lives around precision light manufacturing. When that manufacturing employment declined — from 4,500 workers at peak to fewer than 100 today — it left a workforce culture and a community expectation that manufacturing belongs in Seaford, without the manufacturing jobs that validate that expectation. 4WardCrete is the answer to that gap.

3A. The Displaced Manufacturing Workforce

INVISTA's 2008 restructuring laid off 400 of its 500 remaining Seaford workers. Those workers — and their children — are in their 30s to 50s today. They were trained on precision fiber manufacturing, quality control, shift operations, and light mechanical maintenance. These skills transfer directly to CNC roll-forming operations. The muscle memory of running a production line, meeting dimensional tolerances, and managing material flow is not a skill that retires when the plant shuts down — it is a community asset waiting for a new application.

Sussex County's unemployment rate has historically run above the Delaware state average, driven by the seasonal nature of the beach tourism economy that has partially replaced the manufacturing base. The inland communities — Seaford, Bridgeville, Laurel — do not benefit from tourism employment the way the coastal corridor does. These are the communities where 4WardCrete manufacturing employment lands.

3B. The L.E.E.P. / M4Di Academy Pipeline

L.E.E.P., Inc. (Gwenevere Motley, Executive Director) is already the workforce training and installation partner for the Forward Home program in Wilmington. The L.E.E.P. model — recruiting from communities with limited access to construction careers, providing structured training tied to active job sites, and connecting graduates to employment with the M4D project pipeline — is directly transferable to the Seaford manufacturing context with one important modification: this is manufacturing training, not field installation training.

The M4Di Academy model for Seaford operates at two levels simultaneously:

Track	Description
Track A — Seaford Manufacturing	CNC roll-forming operation, AAC panel staging, container kit assembly. Target recruits: displaced INVISTA/DuPont workforce and their adult children. Prior precision manufacturing experience is an asset, not a prerequisite. Training timeline: 4–6 weeks to basic production competency. Employment: full-time, year-round, salary-based. Not seasonal. Not gig.
Track B — LGS Installation (Field)	Light gauge steel framing installation, AAC panel setting, Allura cladding installation. Target recruits: Wilmington and Sussex County residents with or without prior construction experience. Mirrors the existing L.E.E.P. Forward Home installation training, adapted for LGS + AAC system. Training timeline: 8–12 weeks. Employment: project-based with repeat engagement as Caribbean export pipeline grows.
Track C — M4Di Digital Operations	ALDi Tracker QR labeling, kit inventory management, M4Di OI Tool data entry, installation sequence drawing reading. Entry-level digital operations role that does not require prior construction experience. Target recruits: recent high school or community college graduates. Training timeline: 2–4 weeks with on-the-job continuation. This track is the pipeline into the M4Di predevelopment intelligence platform career path.
Workforce development funding	L.E.E.P. manufacturing training aligns with Delaware Workforce Development Board priorities, USDA Rural Development workforce grants, and EDA workforce programs. Sussex County's designation as a rural development eligible area opens federal workforce training funding that urban Wilmington programs cannot access. The Governor of Delaware meeting (post-grant Q3/Q4 2026) is the vehicle for connecting 4WardCrete manufacturing employment to the DE Skills Compact / vocational pipeline policy.

3C. The Employment Math

Phase 1 factory employment is modest — but the scaling math is compelling:

Phase	Factory jobs	Field / installation	Driver
Phase 1 (Yr 1)	8–12	10–15	Forward Home SIP staging + first Caribbean kit export
Phase 2 (Yr 2)	20–30	20–30	NHT / NaRRA pilot active. Second CNC line operational.
Phase 3 (Yr 3)	40–60	40–60	Commercial building kit line added. Costa Rica distribution node.
At scale (Yr 5)	100–150	100+	Multiple Caribbean markets. US FEMA/HMGP shelter contracts.

THE SEAFORD NARRATIVE: "The world's first nylon plant opened in Seaford in 1939 because DuPont needed precision manufacturing capability near a river port, close to its Wilmington headquarters, and adjacent to a skilled workforce. Eighty-seven years later, Merge4Ward is opening the next manufacturing chapter in Seaford for the same reasons — with a product that the world needs just as urgently as it needed nylon in 1939: resilient, rapidly-deployable, non-combustible housing for communities rebuilding after catastrophic storms. The legacy continues."

3D. The Governor of Delaware — Policy Connection

The planned Governor of Delaware meeting (Q3/Q4 2026, post-grant outcome) is the vehicle for connecting the 4WardCrete manufacturing strategy to the DE Skills Compact and vocational pipeline policy. The pitch is not abstract: it is a real factory, in a real Sussex County community with a real manufacturing workforce legacy, producing a product that the state has an interest in — resilient housing, Caribbean export, workforce development. The Governor's interest in post-Invista economic development for Seaford is documented in Mayor Genshaw's public statements. Merge4Ward is the answer to a problem the Governor already knows exists.

4. The Coastal Construction Case — Why 4WardCrete Belongs Here

The answer is yes across all three coastal threat categories. This section is the technical foundation for every external claim the team makes about 4WardCrete's coastal suitability.

Threat	4WardCrete response	Specification
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Salt air / corrosion	Galvanized LGS framing (G90 for US coastal; G235 for Caribbean HVHZ) provides cathodic protection via sacrificial zinc barrier. AAC enclosure is fully inorganic — immune to salt air corrosion. G90 provides 30–50% longer service life than G60 in moderately corrosive environments.	G90 ASTM A653 minimum for all US coastal 4WardCrete. G235 mandatory for all Caribbean export kits. Epoxy-coated or stainless rebar in reinforced AAC panels. Standard per M4Di OI Tool unit price library.
Hurricane wind loads	LGS framing with engineered hold-downs per AISI S240 is inherently superior to wood frame for wind uplift and racking. CUBiC (Caribbean Uniform Building Code) explicitly recognizes steel framing. ICC 500 hurricane shelter spec: LGS + 200mm reinforced AAC achieves 250 mph wind design in a tested and documented assembly.	AISI S240 design. Hold-down connector schedule per unit. Forward Shelter Kit: ICC 500-2023 compliant. FEMA P-361 eligibility via 200mm reinforced AAC.
Humidity / mold / termites	AAC is inorganic — sand, lime, cement, aluminum powder. Does not rot, mold, or attract termites. Cellular microstructure distributes moisture as disconnected air bubbles rather than connected channels, resisting water penetrability. Acts as seasonal moisture buffer in high-humidity climates.	Exterior Allura fiber cement cladding mandatory for all tropical deployments. Prevents AAC moisture absorption in sustained high-humidity environments. Non-negotiable spec item.
Fire (party wall)	100mm AAC: 3-hr ASTM E119 (inherent). 125mm AAC: 4-hr ASTM E119. Both exceed the 2-hr IRC R302.2 requirement. Non-combustible ASTM E136. AAC will not burn, will not produce toxic fumes, will not contribute to fire spread.	AERCON ESR-1652 (ICC-ES). Forward specification for Type A/B post-Claymont G4. Current: 8" CMU retained for AE24-002.
Delaware coastal / Delmarva zone	Seaford at 115–120 mph ASCE 7 design wind speed. Delaware Bay and Atlantic coast communities at 115–130 mph. 4WardCrete buildings at G90 framing + AAC enclosure outperform wood frame and CMU block at these design wind loads with engineered precision.	ASCE 7-16 / 7-22 wind speed per site zip code. Exposure Category D for coastal sites. Delaware Beach Building Line regulations apply within regulated coastal zones.

5. Implementation Plan — Phases and Triggers

Phase	Window	Key deliverables	Capital / trigger
Phase 0 Foundation	Now→Oct 2026	Contact Framecrete (OEM license terms) Contact AERCON Florida (panel supply + ESR-1652) Seaford facility site study (3 candidates) Product naming + IP strategy Jamaica pilot proposal (4WardCrete spec) 4WardCrete concept drawings (Travis)	<\$25K Existing team capacity NO capital commitment
Phase 1 Pilot	Oct 2026→Mar 2027	Seaford facility lease executed CNC roll-forming equipment ordered 4WardCrete Party Wall spec sheet complete Forward Shelter Kit spec sheet + ICC 500 path L.E.E.P. Seaford Track A training program launched JAMPRO registration completed	\$200K–\$400K Triggered by: NHT/NaRRA MOU OR Wells Fargo Round 3

Phase 2 Production	Mar 2027→Nov 2027	First CNC production run First container kit exported to Jamaica L.E.E.P. Seaford 15–25 workers employed DE Governor briefing on Skills Compact connection FEMA HMGP shelter contract bid (US domestic)	\$500K–\$1.2M Revenue + grant EDA manufacturing grant
Phase 3 Scale	Nov 2027→Nov 2028	Second CNC line operational Costa Rica distribution node Commercial building kit added (heavier gauge) AAC local supply partnership in Caribbean M4Di Academy Seaford: 40–60 workers employed	Revenue-funded + continued grant supplementation

6. Risk Register

Risk	Exposure	Mitigation
INVISTA environmental covenant limits Seaford site re-use	EC prohibits groundwater withdrawal and restricts non-residential use on designated parcels. Not all 750 acres are encumbered, but any acquisition of INVISTA property requires DNREC review of covenant conditions.	Phase 1 targets available light industrial leasehold (1700 Dulaney St / Western Sussex Business Campus) — not the INVISTA site. INVISTA site is long-term ambition only. No commitment before legal/environmental review.
AAC moisture absorption in tropical environments	Unfinished AAC absorbs moisture in sustained high-humidity climates. Reduces thermal performance, accelerates surface degradation over time.	Allura fiber cement cladding is mandatory (not optional) on all exterior-facing AAC in Jamaica and Costa Rica deployments. Same Allura supply chain and L.E.E.P. installation training as Forward Home. Non-negotiable spec item.
LGS corrosion if under-specified	G60 galvanizing in a coastal salt air environment shows accelerated corrosion. If framing is enclosed within AAC panels, corrosion is difficult to detect post-construction.	G90 ASTM A653 minimum for all US coastal 4WardCrete. G235 for Caribbean. Connectors and hold-downs: stainless steel or hot-dipped G235. Document in M4Di OI Tool unit price library as a mandatory line item — not a cost-saving shortcut.
ICC 500 peer review timeline	ICC 500 requires independent structural peer review for all mandated storm shelters. Peer reviews take 2–3 weeks and must be scheduled at 50% CD and 90% CD milestones. This adds time to shelter kit permitting.	Build peer review into the M4Di ALDi Gate 2 and Gate 3 schedule from project initiation. Forward Shelter Kit projects do not reach CD without peer review already scheduled. This is a discipline item, not a cost item.
Framecrete IP conflict	If Framecrete licensing terms restrict OEM AAC sourcing or geographic territory, the independent manufacturing plan is constrained.	Negotiate OEM rights explicitly: right to source AAC independently, right to manufacture LGS framing at Seaford, right to sell in Caribbean territories. Alternative: develop proprietary M4D LGS spec independently using ClarkDietrich ESR-1166P or equivalent — Framecrete does not own LGS technology.

Capital timing vs. revenue	\$200K–\$400K Phase 1 capital before Jamaica revenue is live creates cash flow exposure.	Hard capital trigger: no Seaford lease or CNC commitment before NHT/NaARRA MOU executed or Wells Fargo Round 3 finalist confirmed. Phase 0 desk research costs <\$25K. Discipline enforced by team.
Sussex County workforce skill gap	LGS roll-forming is precision manufacturing. Former INVISTA workers have adjacent skills but not direct experience. Training takes time.	L.E.E.P. Track A program begins at facility lease. 4–6 week training to basic production competency. Former precision manufacturing workers (INVISTA legacy) are the primary recruit pool. Sussex County WDB training grants available.

7. Phase 0 Action Items

Action	What success looks like	By
Contact Framecrete (info@framecrete.com). Position as Caribbean export manufacturer from a Delaware facility — not a franchise prospect. Ask for OEM licensing terms, AAC supply chain details, and whether they have Caribbean territory coverage.	Written response from Framecrete with licensing term framework. If licensing is restrictive, proceed with independent LGS spec (ClarkDietrich ESR-1166P).	30 days
Contact AERCON Florida (Apopka, FL). ESR-1652 panel supply inquiry. Request: 100mm and 200mm wall panel spec sheets, fire test documentation, export pricing for 20-ft and 40-ft container loads to Jamaica and Costa Rica.	AERCON confirmed as viable OEM source with export capability and ESR-1652 documentation. Price per container unit confirmed.	30 days
Develop 4WardCrete concept drawings: LGS + AAC assembly sequence for standard 2BR unit (party wall configuration and shelter kit configuration). Revit + Blender visualization. Not construction documents — design intent and product narrative visuals.	Two sets of visuals: (1) US party wall detail replacing CMU, (2) Jamaica Forward Shelter Kit perspective showing LGS frame + AAC panels + Allura cladding + container unloading sequence.	45 days
Seaford facility research: visit or contact 1700 Dulaney St, Western Sussex Business Campus, and Seaford Business Park (Route 13). Confirm: SF available, asking rate, zoning, 3-phase power availability, loading dock count, ceiling height.	3-facility comparison memo with recommendation. 3-phase power confirmed at preferred site (CNC roll-forming requires it).	45 days
Research Delaware SBIR / USDA Rural Development / EDA Manufacturing grants applicable to Seaford facility. Cross-reference with existing Merge4Ward grant strategy. Identify grant applications executable before Dec 2026.	Grant matrix: 3–5 programs with amount, deadline, eligibility criteria, and owner-team match.	60 days

Develop the 4WardCrete brand narrative for the Jamaica pilot proposal. Made in Delaware. Engineered in America. Reads like concrete to Jamaican homeowners. Installed by a trained local workforce. Not Chinese prefab.	One-page narrative + product comparison talking points for NaRRA / NHT conversations.	60 days
Draft 50-unit Jamaica Forward Shelter Kit pilot proposal for St. Elizabeth recovery zone. Spec: 4WardCrete LGS + AAC. JNBC / CUBiC compliance statement. NHT financing mechanism. L.E.E.P. workforce narrative. ICC 500 reference.	Pilot proposal document ready for NaRRA supplier intake and NHT Private Developer Financing unit conversations.	90 days
Decision gate: proceed to Seaford lease and CNC equipment? Triggered by: NHT/NaRRA MOU executed OR Wells Fargo Round 3 finalist confirmed.	Binary decision with team consensus. No capital commitment without trigger event.	Nov 2026

8. Summary Assessment

Seaford is not an arbitrary location for the 4WardCrete manufacturing operation. It is the most defensible location in the mid-Atlantic for a housing manufacturing business that serves both the US domestic Forward Home market and the Caribbean export market — a community with a precision manufacturing workforce legacy, a rail and port logistics position that no other Sussex County location matches, and a municipal and state government that is actively seeking a successor manufacturing employer for the INVISTA legacy site. The DuPont Nylon Plant opened in Seaford in 1939 because precision manufacturing, a river port, rail access, and a Wilmington connection mattered. All four of those things are still true.

The two proprietary 4WardCrete products are not analogues of Framecrete or any other existing system. They are original Merge4Ward product specifications that use commodity inputs — galvanized steel coil and OEM AAC panels — assembled at Seaford into precisely engineered kits. The IP is in the engineering, the tested assembly documentation, the M4Di predevelopment intelligence layer, and the L.E.E.P. workforce delivery system. The 4WardCrete Party Wall Panel gives the US Forward Home program a non-combustible, 3–4 hour fire-rated party wall at one-third the weight of CMU, without a separate masonry trade. The 4WardCrete Forward Shelter Kit gives the Caribbean reconstruction market a hurricane-resistant, ICC 500-compliant, container-deployable building system that reads as improved concrete to Jamaican code officials and NaRRA procurement officers.

The workforce pipeline completes the value proposition. L.E.E.P.'s manufacturing training track in Seaford is not a new program — it is the same workforce development model that L.E.E.P. already operates in Wilmington, extended to a community where the need is arguably greater and the training base — former INVISTA precision manufacturing workers and their families — is ideally suited to light gauge steel roll-forming production. At scale, 4WardCrete manufacturing in Seaford can employ 100–150 workers. That is not the 4,500-worker peak of the DuPont era — but it is a real, year-round, salary-based manufacturing employment base in a community that lost its anchor employer and has been waiting for what comes next.

The Jamaica pilot proposal is the first external test of the entire thesis. Draft it with 4WardCrete LGS + AAC as the specified system, cite ICC 500 and CUBiC compliance, include the L.E.E.P. workforce narrative, and take it to NaRRA. The response will tell the team everything it needs to know about whether the Caribbean market is ready. Phase 0 costs less than \$25,000 and takes 90 days. That is the price of finding out.

Merge 4 Design × Merge4Ward × M4Di

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4WardCrete™ is a working product name. IP and trademark filing TBD Phase 0.

Phase 0 capital gate:

<\$25K — no Board approval required

Phase 1 capital gate:

\$200K–\$400K triggered by:

**NHT / NaRRR MOU OR Wells Fargo
Rd 3**